

Quantitative Methods for Political Science

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This course introduces undergraduate students to the use and interpretation of quantitative evidence in political science research. Students will learn to think like social scientists: to identify patterns in political outcomes that require explanation, formulate possible explanations for these patterns, and reason critically about claims of cause and effect. In particular, they will learn to ask a central question of empirical social science: under what conditions can we infer that x causes y ? Most political phenomena cannot be studied experimentally because we cannot randomly assign election outcomes, economic crises, regime types, or wars in the same way that we assign treatments in a laboratory. How can we nonetheless use quantitative evidence to better understand the causes and consequences of important political phenomena?

Starting with the foundations of correlation and regression analysis, students will become familiar with both the potential and the pitfalls of the research designs that social scientists use to draw causal inferences from observational data, including fixed effects, differences-in-differences, instrumental variables, and regression discontinuity designs. Throughout the course, we will ask what the appropriate counterfactual is for a causal effect of interest and how best to estimate that counterfactual given the data the world provides us.

Students will also learn to implement these methods themselves. As the rise of generative AI makes the implementation of statistical techniques more straightforward, it becomes more important, not less important, that students understand what a method does and when its use is appropriate. This course therefore places particular emphasis on research design, interpretation, and critical reasoning. The goal of the course is not only to teach students how to use quantitative tools, but to teach them how to use these tools appropriately and to become informed and skeptical readers of quantitative social science. Most importantly, students will learn that the application of sophisticated empirical methods is a complement to, not a substitute for, substantive knowledge and careful reasoning.

Requirements:

Weekly reading: Students are expected to complete the week's readings before the first lecture of the week. The assigned material amounts to 60–80 pages per week.

Quizzes: There will be six in-class quizzes, consisting of multiple-choice and short answer questions drawn from the readings and lectures. Each is worth 5% of the course grade.

Problem sets: There will be four problem sets. Each is worth 5% of the course grade. These assignments provide an opportunity for implementing the empirical tools covered in the course and will build intuition about how different modeling choices change the estimand and produce different results.

Midterm: (Week 8). The midterm will consist of multiple-choice and short answer questions drawn from the readings and lectures. 20% of course grade.

Final exam: The final will consist of multiple-choice and short answer questions drawn from the readings and lectures. 20% of course grade.

Participation: active participation in class based on substantive engagement with the readings is expected and is worth 10% of your course grade. In this course, we will uphold the university's code of conduct and values regarding open intellectual inquiry and equal respect for all class members.

Electronic devices: Laptops are permitted for note-taking and for referring to course readings. Please be conscious of the fact that browsing the internet during class is distracting for the students sitting around you. If you plan to browse the internet during class time, expect to be called upon!

Plagiarism: Plagiarism is a serious breach of the University's code of conduct. It is assumed that you have carefully reviewed the University's policy on plagiarism. Your use of AI must adhere to the university's policy. Please note that you are unlikely to succeed on the midterm and final exam if you have not studied the readings in detail and thought about them yourself.

Accessibility: The University is committed to equal access for students with disabilities. Students with disabilities who have been approved to receive academic accommodations by the University must contact me as soon as possible so that the relevant accommodations can be made.

Week 1: Introduction (2 lectures)

Gerring, John, and Dino Christenson. *Applied Social Science Methodology: An Introductory Guide*. Cambridge: Cambridge University Press, 2017. Ch. 1, "A Unified Framework."

Bueno de Mesquita, Ethan, and Anthony Fowler. *Thinking Clearly with Data: A Guide to Quantitative Reasoning and Analysis*. Princeton: Princeton University Press, 2021. Ch. 1, "Thinking Clearly in a Data-Driven Age."

Week 2: Mapping variation (2 lectures)

Gerring, John. "Mere description." *British Journal of Political Science* 42.4 (2012): 721-746.

Piketty, Thomas, and Emmanuel Saez. "Income inequality in the United States, 1913–1998." *Quarterly Journal of Economics* 118.1 (2003): 1-41 (selections).

Auten, Gerald, and David Splinter. "Top 1 percent income shares: Comparing estimates using tax data." *AEA Papers and Proceedings* 109 (2019): 307-311.

Week 3: Correlation and regression (2 lectures)

Bueno de Mesquita, Ethan, and Anthony Fowler. *Thinking Clearly with Data*. Chs. 2, 4, 5, 6. "Correlation: What Is It and What Is It Good For?" "Correlation Requires Variation," "Regression for Describing and Forecasting," "Samples, Uncertainty, and Statistical Inference."

Week 4: From correlation to causation (2 lectures)

Bueno de Mesquita, Ethan, and Anthony Fowler. *Thinking Clearly with Data*. Ch. 3, 9. "Causation: What Is It and What Is it Good For?" "Why Correlation Doesn't Imply Causation."

Cunningham, Scott. *Causal Inference: The Mixtape*. New Haven: Yale University Press. Ch. 4, "Potential Outcomes Causal Model."

Week 5: Experiments and their limits (2 lectures)

Bueno de Mesquita, Ethan, and Anthony Fowler. *Thinking Clearly with Data*. Ch. 11, "Randomized experiments."

Gerber, Alan S., and Donald P. Green. "The effects of canvassing, telephone calls, and direct mail on voter turnout: A field experiment." *American Political Science Review* 94.3 (2000): 653-663.

Przeworski, Adam. "Is the Science of Comparative Politics Possible?" In *The Oxford Handbook of Comparative Politics*, ed. Carles Boix and Susan Stokes, 147-171. Oxford: Oxford University Press, 2009.

Week 6: Control variables in observational research (2 lectures)

Bueno de Mesquita, Ethan, and Anthony Fowler. *Thinking Clearly with Data*. Ch. 10. "Controlling for Confounders."

Achen, Christopher H. "Let's put garbage-can regressions and garbage-can probits where they belong." *Conflict Management and Peace Science* 22.4 (2005): 327-339.

Lenz, Gabriel, and Alexander Sahn. "Achieving Statistical Significance with Control Variables and Without Transparency." *Political Analysis* 29.3 (2021): 356-369.

Week 7: Fixed effects (2 lectures)

Stock, James, and Mark Watson. *Introduction to Econometrics*. 4th edition. Harlow: Pearson, 2020. Chapter 10, "Regression with Panel Data."

Gamm, Gerald, and Thad Kousser. "Life, literacy, and the pursuit of prosperity: party competition and policy outcomes in 50 states." *American Political Science Review* 115.4 (2021): 1442-1463.

Week 8: Differences-in-differences (2 lectures)

Bueno de Mesquita, Ethan, and Anthony Fowler. *Thinking Clearly with Data*. Ch. 13, "Differences-in-Differences Designs."

García-Montoya, Laura, Ana Arjona, and Matthew Lacombe. "Violence and Voting in the United States: How School Shootings Affect Elections." *American Political Science Review* 116.3 (2022): 807–826.

Hassell, Hans J. G., and John B. Holbein. "Navigating Potential Pitfalls in Difference-in-Differences Designs: Reconciling Conflicting Findings on Mass Shootings' Effect on Electoral Outcomes." *American Political Science Review* 119.1 (2025): 240–260.

Week 9: Instrumental variables (2 lectures)

Cunningham, Scott. *Causal Inference*. Ch. 7, "Instrumental variables."

Acemoglu, Daron, Simon Johnson, and James A. Robinson. "The colonial origins of comparative development: An empirical investigation." *American Economic Review* 91.5 (2001): 1369–1401.

Glaeser, Edward L, Rafael La Porta, Florencio Lopez-de-Silanes, and Andrei Shleifer. "Do Institutions Cause Growth?" *Journal of Economic Growth* 9.3 (2004): 271–303 (selections).

Week 10: Regression discontinuity (2 lectures)

Bueno de Mesquita, Ethan, and Anthony Fowler. *Thinking Clearly with Data*. Ch. 12, "Regression Discontinuity Designs."

DiNardo, John, and David Lee. "Economic Impacts of New Unionization on Private Sector Employers: 1984–2001." *Quarterly Journal of Economics* 119.4 (2004): 1383–1441 (selections).

Frandsen, Brigham. "The Surprising Impacts of Unionization: Evidence from Matched Employer-Employee Data." *Journal of Labor Economics* 39.4 (2021): 861–894 (selections).

Week 11: Exploiting shocks and accidents (2 lectures)

Hopkins, Daniel J., Yotam Margalit, and Omer Solodoch. "Personal economic shocks and public opposition to unauthorized immigration." *British Journal of Political Science* 54.3 (2024): 928–936.

Holman, Mirya R., Jennifer L. Merolla, and Elizabeth J. Zechmeister. "The curious case of Theresa May and the public that did not rally: Gendered reactions to terrorist attacks can cause slumps not bumps." *American Political Science Review* 116.1 (2022): 249–264.

Pahontu, Raluca. "The Democrat Disaster: Hurricane Exposure, Risk Aversion, and Insurance Demand." *The Journal of Politics* 88.1 (2026): 299-315.

Week 12: Exploiting history (2 lectures)

Scheve, Kenneth, and David Stasavage. "Democracy, war, and wealth: lessons from two centuries of inheritance taxation." *American Political Science Review* 106.1 (2012): 81-102.

Zhukov, Yuri, and Roya Talibova. "Stalin's terror and the long-term political effects of mass repression." *Journal of Peace Research* 55.2 (2018): 267-283.

Boix, Carles. "Political emancipation and modern Jewish national identity." *American Political Science Review* 119.4 (2025): 1921-1941.

Week 13: From causal identification to explanation (2 lectures)

Vlandas, Tim, and Daphne Halikiopoulou. "The (forgotten) atomistic fallacy in political science and its implications for how we interpret elections." *European Journal of Political Research* (2026): 1-12.

Williams, Martin J. "Causal Inference, Agency, and the Problem of Inherent Endogeneity." *Annual Review of Political Science* 29 (2026).

Cirone, Alexandra, and Thomas B. Pepinsky. "Historical persistence." *Annual Review of Political Science* 25 (2022): 241-259.